Artificial Intelligence

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Course Outcomes

- **Describe the principal and techniques of Artificial Intelligence including agents, problem-solving, knowledge representation, planning, and learning to gain a comprehensive understanding of intelligent systems.**
- **Use uninformed and heuristic search strategies, and logic-based knowledge representation techniques to solve AI problems in deterministic and adversarial environments.**
- **Apply planning algorithms and learning techniques such as decision trees, reinforcement learning, and statistical learning to build intelligent autonomous agents.**
- **Analyze the effectiveness and suitability of AI search methods, knowledge**

Unit-1

Introduction

- Definition of AI
- History of AI
- Foundations of AI
- Applications of AI

Intelligent Agents

- Agents and Environments
- Good Behavior: The concept of rationality
- The Nature of Environments
- The Structure of Agents
- Problem Solving Agents

Father of A.I.



John McCarthy

Introductio n

- We call ourselves Homo sapiens?

Intelligence-is the ability to learn and understand to solve problems and make decisions

- Understand :How we think?

Perceive, Understand, Predict and Manipulate

- Build Intelligent Entities
- AI is one of the newest field in **Science & Engg.**
- AI started in World War II, name itself was coined in 1956.
- AI not only in Science & Engg and also other areas like Self Driving Cars, Agriculture, Weather Forecasting, Electronic Health Records, Disease Prevention etc.

Introductio n

- AI on the other hand still has openings for full time Scientists.
- AI currently encompasses a huge variety of subfields, ranging from the general (learning and perception) to the specific, such as playing chess, proving mathematical theorems, writing poetry, driving a car on a crowded street, and diagnosing diseases.
- Conclusion:

A.I. is Universal Field.

What is AI

- AI is the theory and development of computer systems capable of performing tasks that historically required human intelligence such as recognizing speech ,making decisions and identifying patterns.
- As a science is to make machines do things that would require intelligence if done by humans.

What is AI

Views of AI fall into four different perspectives --- two dimensions:

1) **Thinking versusActing**
2) **Human versus Rational (which is "easier"?)**

1. **Acting Humanly**	**3. Thinking Rationally**
2. Thinking humanly	**4.Acting Rationally**

Thinking Humanly	**Thinking Rationally**
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985)
"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998)
"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

1.1 Acting humanly: The Turing Test approach



The Turing Test, proposed by Alan Turing (1950) was designed to provide an operational definition of whether or not a computer(machine) can think intelligently like humans"?

The computer would need to possess the following capabilities To pass the total Turing Test:

- **Natural language processing - To enable it to communicate successfully in English.**
- **Knowledge representation - To store what it knows or hears.**
- **Automated reasoning- To use the stored information to answer questions and to draw new conclusions**
- **Machine Learning- To adapt to new circumtances and to detect and extrapolate patterns**
- **computer vision to perceive objects, and**
- **robotics to manipulate objects and move about.**

These six disciplines compose most of AI, and Turing deserves credit for designing a test that remains relevant 60 years later.

1.1.2 Thinking humanly: The cognitive modeling approach

- If we are going to say that a given program thinks like a human, we must have some way of determining how humans think.
- We need to get inside the **actual workings of human minds.** There are three ways to do this.

□**Introspection -** trying to catch our own thoughts.

□**Through psychological experiments -** observing a person in action.

□**Through brain imaging -** observing the brain in action.

Cognitive Science

- The interdisciplinary field of **cognitive science** brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind. Real cognitive science, however, is necessarily based on **experimental investigation** of actual humans or animals.

1.1.3 Thinking rationally: The “laws of thought” approach

- The Greek philosopher **Aristotle** was one of the first to attempt to codify “**right thinking**,” that is, irrefutable reasoning processes.
- His **syllogisms** provided patterns for argument structures that always yielded correct conclusions when given correct premises.

Example:

P1: Rahul is a man
P2: All men are mortal
Conclusion: Rahul is mortal

1.1.3 Thinking rationally: The “laws of thought” approach

- These laws of thought were supposed to govern the operation of the mind; their study initiated the field called **logic**.
- The so-called **logicist tradition** within artificial intelligence hopes to build on such programs to create intelligent systems.

There are two main **obstacles** to this approach

- Informal knowledge
- Big difference between solving a problem “in principle” and solving it in practice.

1.1.4 Acting rationally: The Rational agent approach

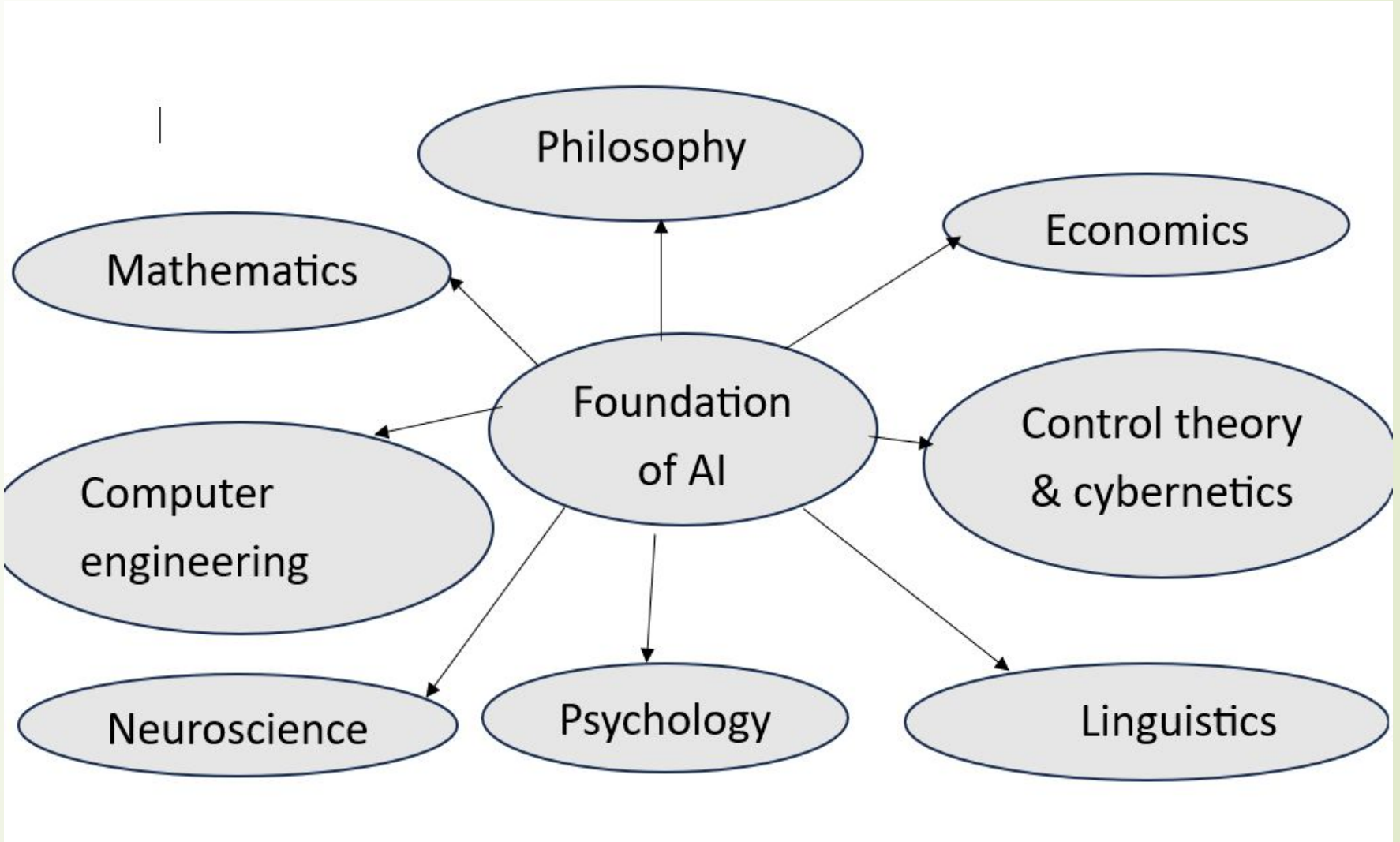
- An **agent** is just something that acts.
- A **rational agent** is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.
- Making **correct inferences** is sometimes part of being a rational agent, because one way to act rationally is to reason logically to the conclusion that a given action will achieve one's goals and then to act on that conclusion.

Example: Recoiling from a hot stove is a reflex action.

- All the skills needed for the Turing Test also allow an agent to act rationally. Knowledge representation and reasoning enable agents to reach good decisions.
- The rational-agent approach has two advantages over the other approaches. 1.laws of thought, 2. human behavior or human thought.

1.2 FOUNDATIONS OF AI

- Philosophy
- Mathematics
- Economics
- Neuroscience
- Psychology
- Computer engineering
- Control theory and cybernetics
- Linguistics


Philosophy
Economics
Mathematics
Foundation of AI
Control theory & cybernetics
Computer engineering
Neuroscience
Psychology
Linguistics

Philosophy

- Can **formal** rules be used to draw valid conclusions?
- How does the **mind** arise from a physical **brain**?
- Where does **knowledge** come from?
- How does knowledge lead to **action**?

Reasoning

- Rationalism
- Dualism
- materialism
- Empiricism of syllogisms for proper Reasoning
- Induction
- Logical positivism and Observation sentences
- Confirmation Theory.

Rationalism

- Rationalism is a method of thinking that is marked by being a deductive and abstract way of reasoning.

Example: Slavery is wrong.

In the context of rationalism, this conclusion might come from applying universal moral principles through logical reasoning. For instance:

Premise 1: All humans have inherent dignity and equal rights.

Premise 2: Slavery treats some humans as property, denying their rights and dignity.

Dualism

- part of the human mind (or soul or spirit) that is outside of nature, exempt from physical laws. Animals, on the other hand, did not possess this dual quality; they could be treated as machines.
- **Materialism:** It holds that the brain's operation according to the laws of physics constitutes the mind.

Empiricism

- Empiricism in the philosophy of science emphasizes **evidence**, especially as discovered in experiments.

Logical positivism and Observation sentences

- It holds that all knowledge can be characterized by logical theories connected, ultimately, to observation sentences that correspond to sensory inputs; thus logical positivism combines rationalism and empiricism

Confirmation Theory

- To analyze the acquisition of knowledge from experience.

Mathematics

- Logic
- Computation
- Tractability
- Incompleteness theorem
- NP-completeness
- Probability

Logic

- Logic in computer science (or computational logic) is that branch of mathematical logic which is approximately the intersection between mathematical logic and computer science

Computation(Theory of Computation)

- Traffic Lights, Video Games, CPU Controllers, Protocol Analysis, Regular Expression Matching, Vending Machines, Speech Recognition, Natural Language Processing.

Computation

- Regular Expression = 0(10)*

Tractability

- A Problem is called intractable if the time required to solve instances of the problem grows exponentially with the size of the instances.
- So-called easy, or tractable, problems can be solved by computer algorithms that run in polynomial time; i.e., for a problem of size n, the time or number of steps needed to find the solution is a polynomial function of n.

Examples:

Searching an unordered list, Searching an ordered list, Sorting a list

Incompleteness theorem

- The first incompleteness theorem states that in any consistent formal system within which a certain amount of arithmetic can be carried out, there are statements of the language of which can neither be proved nor disproved.

NP-Completeness

- In computational complexity theory, a problem is NP-complete when: It is a decision problem, meaning that for any input to the problem, the output is either "yes" or "no". When the answer is "yes", this can be demonstrated through the existence solution.

NP-Completeness

Example:

- The traveling salesman problem, satisfiability problems, and graph-covering problems.

Traveling salesman problem

- Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?
- The traveling salesman problem (TSP) is an algorithmic problem tasked with finding the shortest route between a set of points and locations that must be visited.

Probability

- When we flip a coin a very large number of times, we find that we get half heads, and half tails.
- We conclude that the probability to flip a head is 1/2, and the probability to flip a tail is 1/2.

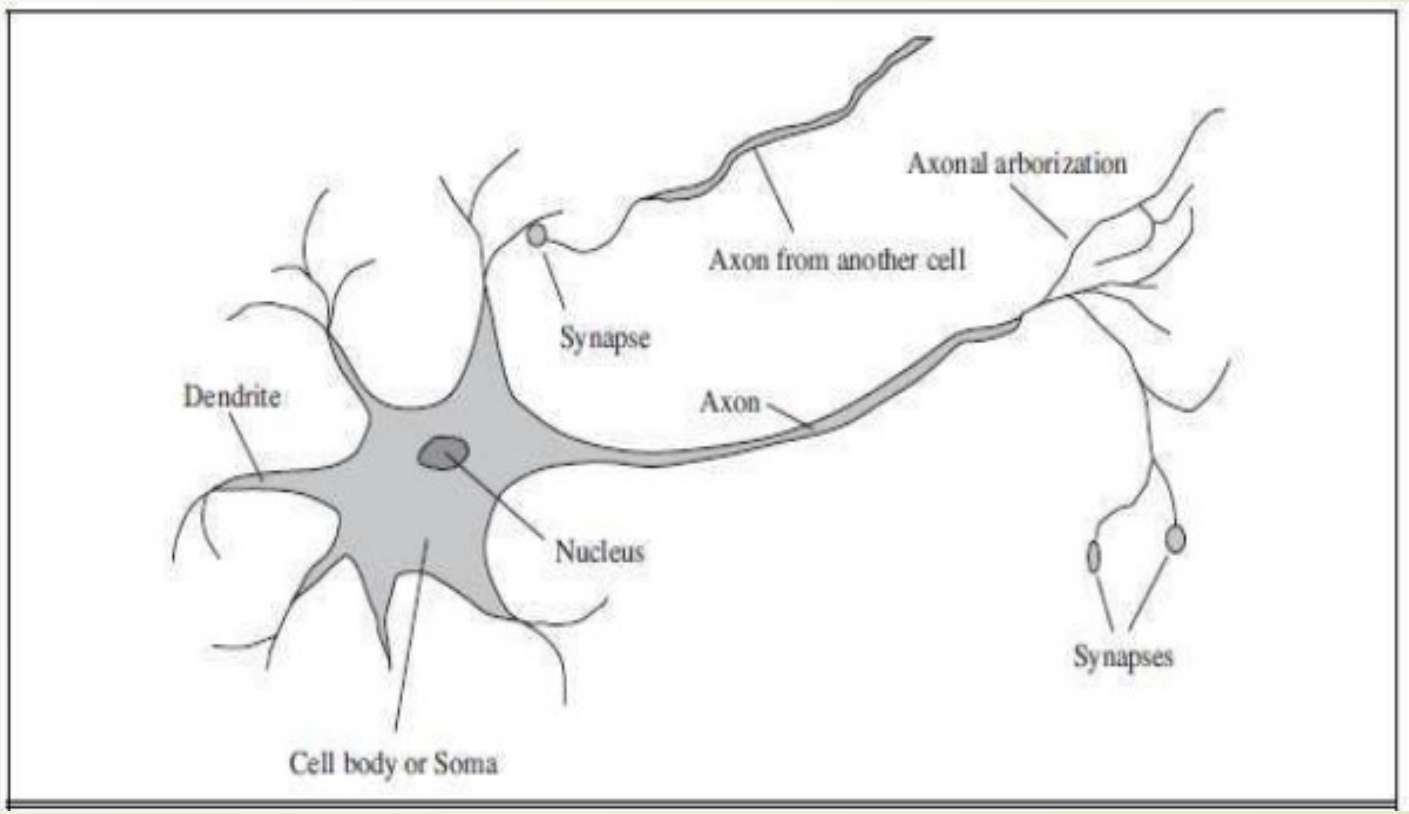
Economics

- Most people think of economics as being about money, but economists will say that they are really studying how people make choices that lead to preferred outcomes or utility.

- **Decision theory:** When McDonald's offers a hamburger for a dollar, they are asserting that they would prefer the dollar and hoping that customers will prefer the hamburger.
- Decision theory, which combines probability theory with utility theory, provides a formal and complete framework for decisions (economic or otherwise) made under uncertainty—that is, in cases where probabilistic descriptions appropriately capture the decision maker's environment.
- **Game Theory:** For "small" economies, the situation is much more like a game: the actions of one player can significantly affect the utility of another (either positively or negatively). Unlike decision theory, game theory does not offer an unambiguous prescription for selecting actions

Neuroscience

- How do brains process information?
- Neuroscience is the study of the nervous system, particularly the brain.
- Brain consisted of nerve cells, or neurons.



Neuroscience

- The parts of a nerve cell or neuron. Each neuron consists of a cell body, or **soma**, that contains a cell nucleus. Branching out from the cell body are a number of fibers called **dendrites** and a single long fiber called the **axon**. The axon stretches out for a long distance, much longer than the scale in this diagram indicates.
- Typically, an axon is 1cm long (100 times the diameter of the cell body), but can reach up to 1 meter.
- A neuron makes connections with 10 to 100,000 other neurons at junctions called **synapses**.
- Signals are propagated neuron to neuron by a complicated

Neuroscience

- The signals control brain activity in the short term and also enable long-term changes in the connectivity of neurons. These mechanisms are thought to form the basis for learning in the brain. Most information processing goes on in the **cerebral cortex**, the outer layer of the brain. The basic organizational unit appears to be a column of tissue about **0.5 mm** in diameter, containing about 20,000 neurons and extending the full depth of the cortex about 4 mm in humans).

Psychology

- How do humans and animals think and act?
- Behaviorism
- **Cognitive psychology:** Views the brain as an information-processing device
- **Cognitive science:** Cognitive science is the interdisciplinary study of the mind and intelligence, combining knowledge and methods from multiple fields to understand how humans (and machines) think, learn, perceive, remember, and make decisions.
- For artificial intelligence to succeed, we need two things: intelligence and an artifact.

Computer Science

- To build an efficient computer for AI, we need both intelligence and a powerful artifact—the computer itself.
- Modern computers evolved rapidly from early wartime inventions like the Heath Robinson, Colossus, and ENIAC, improving in speed, capacity, and affordability over time. Early pioneers like Konrad Zuse and Charles Babbage laid the foundations, while Ada Lovelace envisioned computers doing creative tasks.
- AI has benefited from and contributed to computer science, influencing areas like interactive systems, programming paradigms, and memory management—making today's AI-capable computers both fast and smart.

Control theory and cybernetics

- How can artifacts operate under their own control?
- **Control theory:** Control theory is a field of engineering and mathematics that deals with how to influence the behavior of systems using feedback.

  Eg: robotics, autonomous vehicles, and real-time decision-making systems
- **Homeostatic devices:** containing appropriate feedback loops to achieve stable adaptive behavior.
- **Objective function:** Modern control theory, especially the branch known as stochastic optimal control, has as its goal the design of systems that behave optimally, and that maximize an objective function over time.

Linguistics

- How does language relate to thought?

Computational linguistics or natural language processing.

- This is a hybrid field combining linguistics and computer science that focuses on enabling computers to understand, interpret, and generate human language. It emerged when researchers realized that understanding language involves more than just grammar—it requires context, meaning, and world knowledge. NLP handles tasks like speech recognition, translation, and text analysis.

Linguistics

Knowledge representation

- This refers to how information and facts about the world are stored in a computer so it can reason and understand like a human. In language understanding, knowledge representation helps computers make sense of words by linking them to real-world concepts and context—going beyond sentence structure to meaning.

1.3 THE HISTORY OF AI

- The gestation of artificial intelligence (1943–1955)
- The birth of artificial intelligence (1956)
- Early enthusiasm, great expectations (1952– 1969)
- A dose of reality (1966–1973)
- Knowledge-based systems: The key to power? (1969–1979)
- AI becomes an industry (1980–present)
- The return of neural networks (1986–present)
- AI adopts the scientific method (1987–present)
- The emergence of intelligent agents (1995– present)
- The availability of very large data sets (2001– present)

Gestation of Artificial Intelligence (1943-1955)

- **Year 1943:** The first work which is now recognized as AI was done by Warren McCulloch and Walter pits in 1943. They proposed a model of artificial neurons.
- **Year 1949:** Donald Hebb demonstrated an updating rule for modifying the connection strength between neurons. His rule is now called Hebbian learning.
- **Year 1950:** The Alan Turing who was an English mathematician and pioneered Machine learning in 1950. Alan Turing publishes "Computing Machinery and Intelligence" in which he proposed a test. The test can check the machine's ability to exhibit intelligent behavior

The birth of Artificial Intelligence (1956)

- **Year 1955:** An Allen Newell and Herbert A. Simon created the "first artificial intelligence program“ Which was named as "Logic Theorist". This program had proved 38 of 52 Mathematics theorems, and find new and more elegant proofs for some theorems.
- **Year 1956:** The word "Artificial Intelligence" first adopted by American Computer scientist John McCarthy at the Dartmouth Conference. For the first time, AI coined as an academic field.

Early enthusiasm, great expectations (1952–1969)

- In AI's early years, limited computing power made even simple intelligent behavior seem remarkable. Many believed machines couldn't perform certain task. Newell and Simon’s early success was followed up with the General Problem Solver, or GPS. Unlike Logic Theorist, this program was designed from the start to imitate human problem-solving protocols.
- In 1958, John McCarthy created Lisp, introduced the Advice Taker for general AI, helped invent time sharing, and later advanced logic-based AI at Stanford, while Minsky focused on practical, anti-logic approaches.

Early enthusiasm, great expectations (1952–1969)

- Minsky supervised a series of students who chose limited problems that appeared to require intelligence to solve. These limited domains became known as **microworlds**.

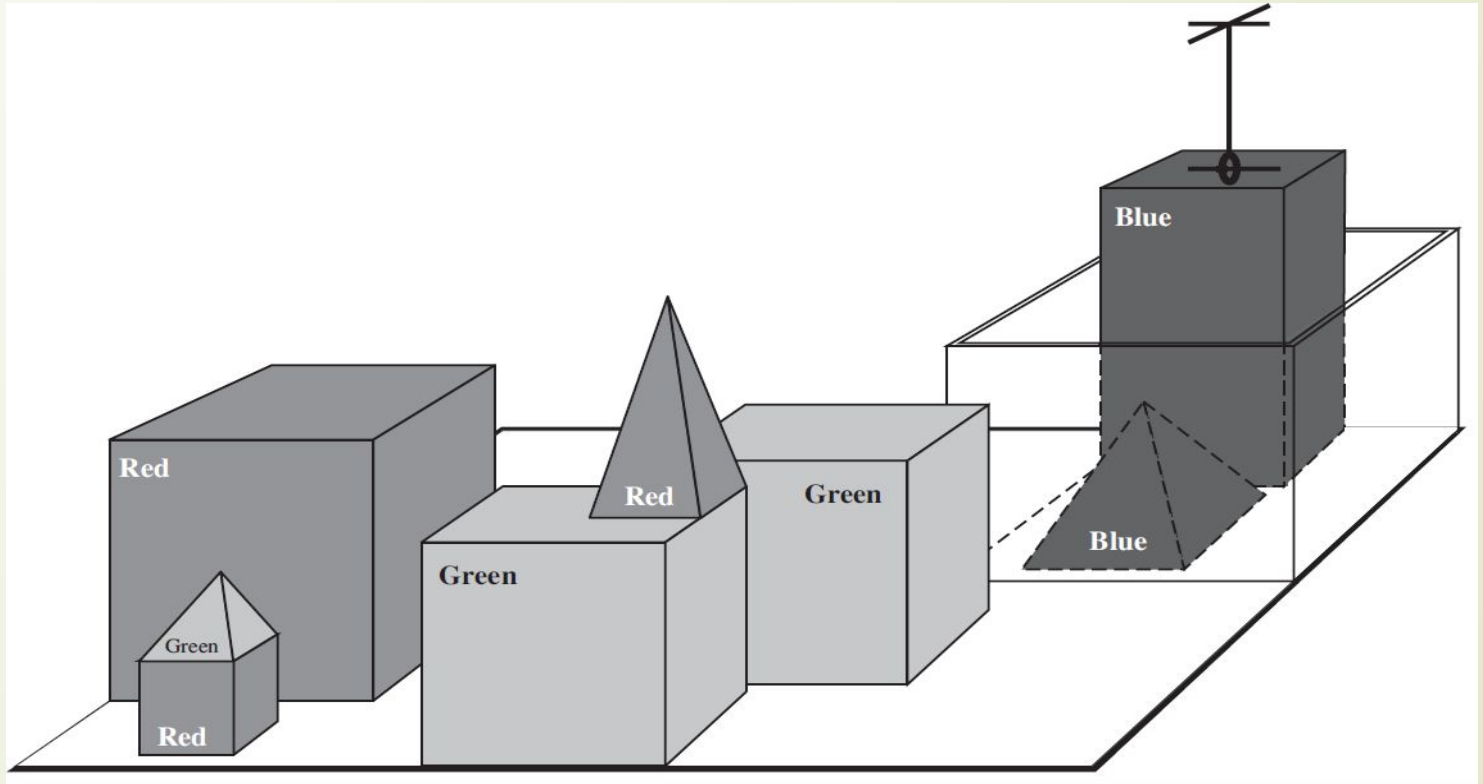


Figure 1.4 A scene from the blocks world. SHRDLU (Winograd, 1972) has just completed the command "Find a block which is taller than the one you are holding and put it in the box."

Early enthusiasm, great expectations (1952–1969)

- The work of Winograd and Cowan (1963) showed how a large number of elements could collectively represent an individual concept, with a corresponding increase in robustness and parallelism

A Dose Of Reality (1966–1973)

- In 1957, Herbert Simon boldly predicted that machines would soon match human intelligence in thinking, learning, and problem-solving.
- Early AI predictions, like Simon's, were overly optimistic due to initial successes on simple tasks. However, systems failed on real-world problems due to lack of background knowledge and computational complexity, as seen in flawed machine translation and slow theorem proving.
- In 1966, a report by an advisory committee found that "there has been no machine translation of general scientific text, and none is in immediate prospect."

A Dose Of Reality (1966–1973)

- Early systems turned out to fail miserably when tried out on wider selections of problems and on more difficult problems

i) Early programs knew nothing of their subject matter; they succeeded by means of simple syntactic manipulations

ii) intractability of many of the problems that AI was attempting to solve.

iii) some fundamental limitations on the basic structures being used to generate intelligent behavior.

A Dose Of Reality (1966–1973)

1969: Minsky and Papert's Perceptron's book showed limitations of simple neural networks, stalling neural-net research. Backpropagation algorithms for multilayer networks were discovered (Bryson and Ho) but gained attention only decades later.

1973: The Lighthill Report criticized AI for combinatorial explosion; UK cut most AI research funding, limiting it to two universities.

Knowledge-based systems: The key to power? (1969–1979)

- Early AI used weak, general-purpose search methods that struggled with complex problems.
- The DENDRAL program, made in 1969 at Stanford, was an example of this better way. It helped scientists figure out the shape of molecules using data from a machine called a mass spectrometer. At first, the program tried all possible shapes, but that took too long.
- Later, it copied how real chemists think. It looked for familiar patterns in the data that matched parts of known molecules. This

Knowledge-based systems: The key to power? (1969–1979)

- For example, the following rule is used to recognize a ketone (C=O) subgroup (which weighs 28):
- if there are two peaks at x1 and x2 such that
- (a) x1 + x2 = M + 28 (M is the mass of the whole molecule);
- (b) x1 − 28 is a high peak;
- (c) x2 − 28 is a high peak;
- (d) At least one of x1 and x2 is high.
- then there is a ketone subgroup

Knowledge-based systems: The key to power? (1969–1979)

- Building on DENDRAL's success, Feigenbaum and his team at Stanford created the Heuristic Programming Project (HPP) to apply **expert systems** to other fields. Their next big system was MYCIN, which diagnosed blood infections using around 450 expert rules. It matched expert-level performance and outperformed junior doctors.

Unlike DENDRAL, MYCIN's rules:

- 1. Came from interviews with medical experts, not from theory.
- 2. Handled uncertainty using a system called certainty factors, to

Knowledge-based systems: The key to power? (1969–1979)

- 1975 – Minsky: Introduced frames for structured knowledge.
- 1976 – Rieger: Modeled human memory for language.
- 1977 – Schank & Abelson: Developed scripts for stereotypical events.
- 1978 – Wilensky: Studied goals and plans in language.
- 1981 – Schank & Riesbeck: Focused on memory-based language understanding.
- 1983 – Dyer: Simulated emotion and memory in story understanding.

AI becomes an industry (1980–present)

- The first successful expert system, R1, was used by Digital Equipment Corporation in 1982 to help configure computers, saving millions of dollars. By the late 1980s, many companies like DEC and DuPont were using expert systems, and nearly every big U.S. company had an AI team.
- The duration between years 1974 to 1980 was the first AI winter duration. AI winter refers to the time period where computer scientist dealt with a severe shortage of funding from government for AI researches.

The return of neural networks (1986–present)

- In the mid-1980s, researchers rediscovered the backpropagation algorithm (first created in 1969) and used it to train neural networks. This sparked interest in connectionist models—systems that learn like the brain by connecting many small units (neurons).
- These models were seen as rivals to symbolic AI, which used logic and rules. Some researchers argued that real thinking might not involve symbols at all. Today, most agree that both approaches—connectionist and symbolic—can work together.

The return of neural networks (1986–present)

- Neural network research has since split into two areas:

  i) Building powerful learning models

  ii) Studying how real brains work

AI adopts the scientific method (1987–present)

- AI research focuses on real-world applications, using rigorous experiments and shared data. Technologies like speech recognition and machine translation now rely on solid models like Hidden Markov Models (HMMs) and statistical methods. There are two aspects in HMM

  i) rigorous mathematical theory

  ii) process of training on a large corpus of real speech data.

- Neural networks, once experimental in the 1980s, are now mature and often combined with techniques from data mining and

AI adopts the scientific method (1987–present)

- Judea Pearl's book in 1988 made Bayesian networks popular in AI for reasoning with uncertainty, combining ideas from classical AI and neural networks.
- This marked a shift toward probability and decision theory, supported earlier by Peter Cheeseman (1985).The idea of normative expert systems—which act rationally using decision theory—was promoted by Pearl, Horvitz, and Heckerman. Fields like robotics, computer vision, and knowledge representation have also advanced by using machine learning, reversing earlier isolation from mainstream AI.

The emergence of intelligent agents (1995- present)

- Researchers are now focusing on building complete intelligent agents, not just solving isolated AI problems. The SOAR (State,Operator,And,Result) architecture (Laird et al., 1987; Newell, 1990) is a major example. AI is now widely used on the Internet, powering bots, search engines, and recommender systems. This shift led to:

i) Better handling of uncertainty from imperfect sensors.

ii) Closer ties between AI and fields like control theory and economics.

iii) Successes in areas like robotic car control, using a mix of sensing

The emergence of intelligent agents (1995- present)

- However, AI pioneers like McCarthy, Minsky, and Nilsson expressed concern that modern AI focuses too much on narrow tasks. They promote Human-Level AI (HLAI)—machines that truly think and learn—with the first symposium in 2004.
- A related area, Artificial General Intelligence (AGI), aims for systems that can learn and act in any environment. It began with ideas from Ray Solomonoff (1964) and held its first AGI conference in 2008.

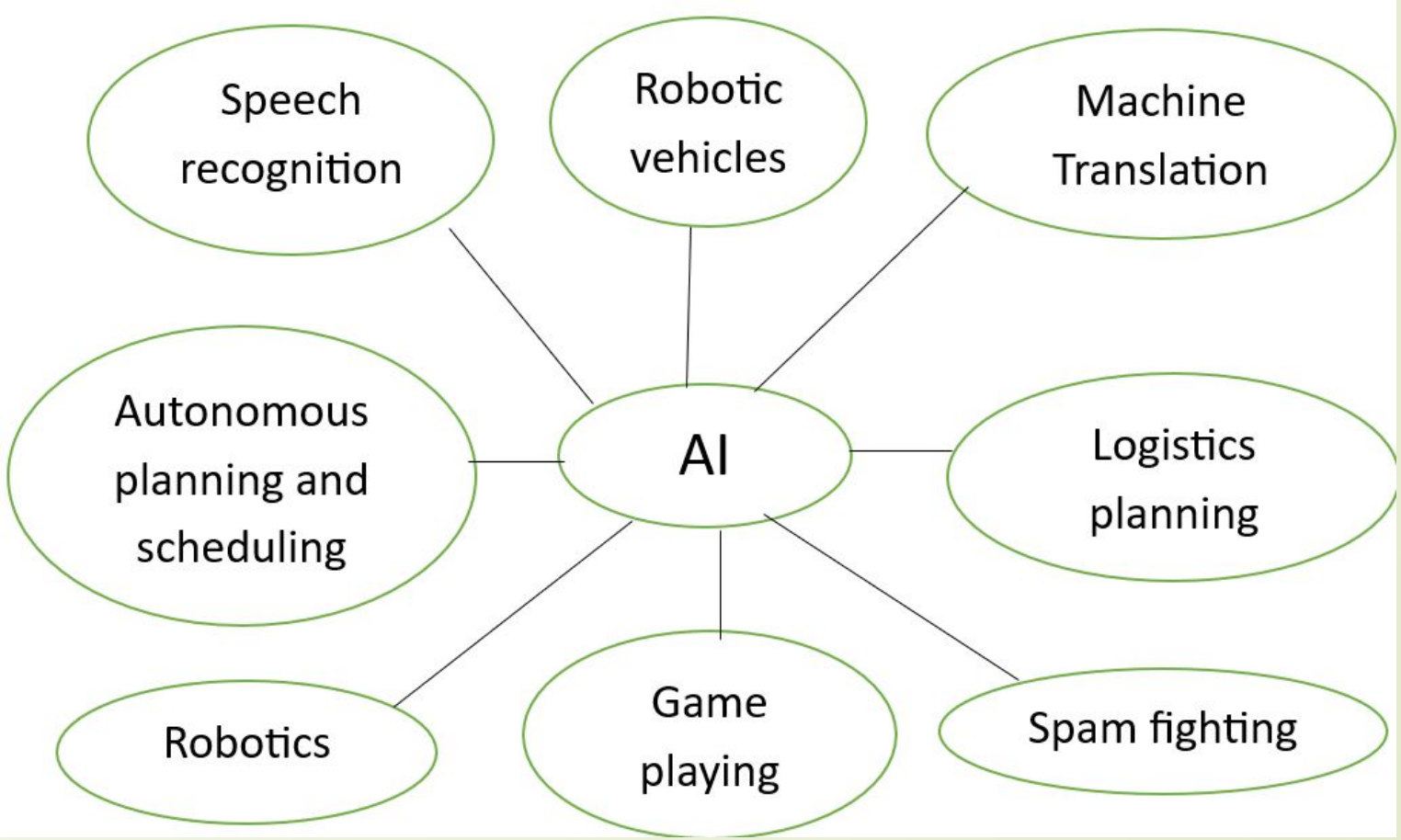
The availability of very large data sets (2001–present)

- 1995 – Yarowsky: Solved word meaning problems (like "plant" = tree or factory) using just definitions and large amounts of unlabeled text—no need for human-labeled examples.
- 2001 – Banko & Brill: Showed that more data improves results more than better algorithms; even simple algorithms work great with huge datasets.
- 2007 – Hays & Efros: Developed a method to fill missing parts in photos; performance jumped when dataset size grew from 10,000 to 2 million images.

The availability of very large data sets (2001–present)

- 2009 – Halevy et al.: Argued that AI's knowledge bottleneck can be solved through machine learning with lots of data, not hand-coded knowledge.
- 2005 – Kurzweil & Havenstein: Noted that thousands of AI tools were now used across industries, suggesting a new AI Spring after the past "AI Winter."

1.4 Applications of AI



1. Robotic vehicles

- A driverless robotic car named STANLEY sped through the rough terrain of the Mojave dessert at 22 mph, finishing the 132-mile course first to win the 2005 DARPA Grand Challenge. STANLEY is a Volkswagen Touareg outfitted with cameras, radar, and laser rangefinders to sense the environment and onboard software to command the steering, braking, and acceleration (Thrun, 2006).
- The following year CMU's BOSS won the Urban Challenge, safely driving in traffic through the streets of a closed Air Force base, obeying traffic rules and avoiding pedestrians and other vehicles.

2. Speech recognition

- A traveler calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.

3. Autonomous planning and scheduling

- A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft (Jonsson et al., 2000). REMOTE AGENT generated plans from high-level goals specified from the ground and monitored the execution of those plans—detecting, diagnosing, and recovering from problems as they occurred.
- Successor program MAPGEN (Al-Chang et al., 2004) plans the daily operations for NASA's Mars Exploration Rovers, and MEXAR2 (Cesta et al., 2007) did

4. Game playing

- IBM's DEEP BLUE became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match (Goodman and Keene, 1997). Kasparov said that he felt a "new kind of intelligence" across the board from him.
- Newsweek magazine described the match as "The brain's last stand." The value of IBM's stock increased by $18 billion. Human champions studied Kasparov's loss and were able to draw a few matches in subsequent years, but the most recent human-computer matches have been won convincingly by the computer.

5. Spam fighting

- Each day, learning algorithms classify over a billion messages as spam, saving the recipient from having to waste time deleting what, for many users, could comprise 80% or 90% of all messages, if not classified away by algorithms.
- Because the spammers are continually updating their tactics, it is difficult for a static programmed approach to keep up, and learning algorithms work best (Sahami et al., 1998; Goodman and Heckerman, 2004).

6. Logistics planning

- During the Persian Gulf crisis of 1991, U.S. forces deployed a Dynamic Analysis and Replanning Tool, DART (Cross and Walker, 1994), to do automated logistics planning and scheduling for transportation.

- This involved up to 50,000 vehicles, cargo, and people at a time, and had to account for starting points, destinations, routes, and conflict resolution among all parameters.
- The AI planning techniques generated in hours a plan that would have taken weeks with older methods. The Defense Advanced Research Project Agency (DARPA) stated that this single application more than paid back DARPA's 30-year investment in AI.
- The iRobot Corporation has sold over two million Roomba robotic vacuum cleaners for home use.
- The company also deploys the more rugged PackBot to Iraq and Afghanistan, where it is used to handle hazardous materials, clear explosives, and identify the location of snipers.

7. Robotics

8. Machine Translation

- Computer program automatically translates from Arabic to English, allowing an English speaker to see the headline “Ardogan Confirms That Turkey Would Not Accept Any Pressure, Urging Them to Recognize Cyprus.”
- The program uses a statistical model built from examples of Arabic-to-English translations and from examples of English text totaling two trillion words (Brants et al., 2007). None of the computer scientists on the team speak Arabic, but they do understand statistics and machine learning algorithms.